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Cognitive IoT system with intelligence techniques in sustainable computing environment

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ABSTRACT

Forest border crossing animals creates major societal related issues, in addition to endangering their own lives. This is the objective focused in this paper targeting the species "The Elephant", incorporating with technical methodologies namely, multi-sensor data fusion, cognition theories and computational intelligence techniques. Multi-sensor data fusion provides three level detection of target, along with its related outputs, which improves performance metrics. Cognition theory resulted in obtaining other interesting features about the target. Computational intelligence techniques integrate and conclude the presence of the target in the pseudo-boundary. The technical combination enhances the novelty of the research work, resulting in achieving remarkable accuracy and minimized false alert. An IoT kit was designed and deployed in the real time wild environment in Hosur forest region for collecting the data of Elephant. Further, the notification is sent to the registered mobile of the forest authority, as an early warning for chasing the pachyderm back to the forest.

1. Introduction

Boundary Intellect provides monitoring capability to observe border areas, where freedom of movement is restricted. The facts are simple, straightforward, well-known but unsettling. Now-a-days technology is used increasingly at the physical borders. A sensor provides an intelligent solution for continuous monitoring of landscapes effectively. Wireless Sensor Network (WSN) is classified into Wireless Underground Sensor Network (WUSN) and Wireless Aboveground Sensor Network (WASN) [1]. This research work tries to utilize the advantage of both WUSN and WASN. Multi-sensor systems afford a determined depiction of the surroundings that a single sensor cannot provide. Multi-sensor fusion technique can integrate a large amount of data and knowledge representing the same environment and obtain a steady, exact and valuable representation of that environment. According to Boundary monitoring research the main role is played by spatio temporal data [2]. These kinds of data are also known as moving object database or real-time locating system, which exactly deals with space and time information [3]. WSN is well suited for collecting spatio-temporal data. These data may be independent or redundant and can be collected from various sensors, may be at the same or different time, which highly increases the profit of information in comparison with that from a single sensor.

The two different methodologies of Boundary Intellect include Lagrangian methodology and Eulerian methodology. Lagrangian methodology is based on specific individual, which entails following that individual using Radio Frequency Identification (RFID), Global Positioning System (GPS), Inertial sensors and Radio transmitters etc. [4–8]. Eulerian methodology is based on place and contracts with the existence of an individual or a group in that location and the changes occurring over time using Radar, Camera, thermometer, PIR, thermal camera, electric nose, microphone and geophone etc. [9,10]. This paper focuses on Eulerian based approach for detecting the target entering or crossing the restricted border area. The target is specifically focused as elephant. Some of the researchers explained their algorithm for acquisition bandwidth [11].

The inclusion of Cognition theory enhances the novelty of the research work. Boundary Intellect merges the suitable combination of target detection along with analyzing the behavior of the target utilizing cognition [12]. The theory of cognition deals with analyzing the mental processes primarily, before further procedures. Cognition is the act of obtaining knowledge through discrimination, intellectual, imaginings, retention, arbitrating, problem-solving, and discerning attention. Computational intelligence plays a major role in Boundary Intellect, because sensors data are uncertain and incomplete. Hence a

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general framework for integrating these uncertain measurements can be adopted by utilizing computational intelligence techniques [13]. The multi-sensor integration detection system serves as an aid to appraise the trustworthiness of the core performance metrics namely accuracy and fault tolerance [14]. This paper is novel from [12], because of its implementation and design of real time IoT setup. The experiments were conducted in different environmental conditions and performance metrics were evaluated. The performance metrics focussed in this research are detection rate and false alert. These two parameters must be significant in any of the elephant detector systems. The increase in detection rate leads to decrease in false alert. The main intention is to provide an early warning to the forest officials and the people residing near forest border and to provide a long term elephant detector system which works under all climatic conditions, with high accuracy and minimized false alert. This paper attempts to create an integrated Boundary Intellect system utilizing the benefits of different sensors incorporating with computational intelligence and cognition theory.

2. Related work

The literature survey of this paper incorporates three different areas, namely, acoustic, image and vibration of elephant. Many existing works were implemented for elephant detection but resulted in short term solution because of multiple reasons; one significant issue is climatic changes. Climatic conditions play a vital role in any detector system. Role of sensors varies depending on the nature of the environment. This is the major drawback faced by existing elephant detector systems.

Acoustic detectors [15,16] and [17] states that detection of elephants utilizing the vocalization is an efficient way of recognition. The communications in elephants are quiet interesting, low frequency sounds were utilized for long distance message transfer. Hence presence of elephant can be identified even in long distance using acoustic sensor. But the sensor collects both positive and negative sound samples. Negative sound samples include other animal and bird sounds, wind, rain and vehicle sounds. Even the environmental variations highly affect the performance of acoustic sensors. Because sound and speed depend on the environmental factors such as temperature, water vapor, atmospheric pressure and CO_2 content in the air, hence affects the accuracy of elephant vocalization. This sensor provides out of sight recognition and capable of collecting very minute frequency sounds.

Vibration detectors [3,18] and [19] briefs the elephant vibration detection. Recording of elephant vibrations generated creates different pattern and can be easily distinguishable from other wild. This sensor is highly advantageous in intruder detection systems because it does not directly affect the target and its oblivious to know the existence. The vibration detector is more efficient than other two sensors because physical damage on the system is the significant factor which leads to short term solution. Physical damage is completely avoided due to its oblivious nature. But the environmental variations in the soil humidity affect the system.

Image detector [17] and [9] is a successful technique of identifying elephant. Image generation creates a visual proof about the presence of elephant and many researchers followed camera traps for recording of images and videos in wild environment. Most advancements in image detector using deep learning convolutional neural networks are utilized in this paper for effective recognition of elephants in wild environment. These detector systems have proved to be effective in elephant recognition but do not provide long term solution. This paper attempts to provide long term solution by introducing integration and computation intelligence technique.

3. Architectural taxonomy

The big picture of this paper includes three units of Boundary Intellect system as shown in Fig. 1.

This research work is an attempt for identifying the target (designed for only one species — "The Elephant") entering/crossing the boundary, along with its cognitive features. The focused performance metrics accuracy and false alert, is achieved by incorporating recent technical advancements which is discussed in following sections.

Unit 1 — **Multi-sensor data** module is an effective way for optimum utilization of large volume of data from multiple sources. This module consists of three sensors specifically, camera, vibration and acoustic.

- Camera provides on-sight detection of Elephant entering/crossing the boundary.
- Vibration sensor and acoustic sensor provides out-of-sight detection.

Unit 2 — **Cognition module** extracts different properties and behavior of Elephant from individual sensor output, by predicting various intentions and reasoning utilizing "Theory of Mind".

Unit 3 — **Integration module** is the output module which detects the presence of Elephant in the pseudo-boundary [12] of Boundary Intellect system using Fuzzy Cognitive Map and Fuzzy rule extraction method.

4. Experimental design and analysis

4.1. Multi sensor data unit

The objective of this unit is to avoid the limitations of single sensor and to improve the accuracy of target detection. The Multi Sensor data unit includes three different sensors as described in the architecture diagram (Fig. 1). The choice of sensors is based on the literature survey of successfully proven target detection sensors in various fields. The descriptions of these sensors are categorized into three sections.

- Visual sensor
- Vibration sensor
- Acoustic sensor

During November 2014 to September 2017, the dataset of elephants utilizing all three sensors were collected on periodic basis from different areas based on their migration nature. The areas focused were Hosur, Bannerghatta, Shimoga (Sakrebailu) and Dubare (a combination of Tamilnadu and Karnataka forest border areas). Limited recordings were composed from Hosur and Bannerghatta. The rest were collected from elephant camps, which last throughout the year (Dhubare and Shimoga) [20].

The significant factor focussed in this research is distance metric in the sensor unit, which is categorized into three namely, SHORT (S), MEDIUM (M) and LONG (L). Depending on the number of sensors or devices embedded with sensors, the coverage was limited. For example, visual detection performed by one camera is limited to its field view as shown in Fig. 2. For more accuracy, multiple camera results should be combined. This section describes about the difference in the coverage and calculation of distance metric about each sensor involved in the experimental setup.

4.1.1. Visual sensor

The visual sensor used for elephant detection is Canon XC10 camera. The visual classification happens based on triggering mechanism when any object enters the fixed boundary. Boundary Sense Deep Learning (BSDL) algorithm [14] is proposed for identifying the captured image accurately with technical advancements. The dataset incorporating various images of elephants postures collected from different areas under different environments. BSDL Algorithm consists of three steps. First step compares the raw image with the query image, which is in the background dataset (incorporating only background images). If the raw image consists of any object, then it is moved to next module, otherwise dropped. Second step includes Background Subtraction (BS) technique,

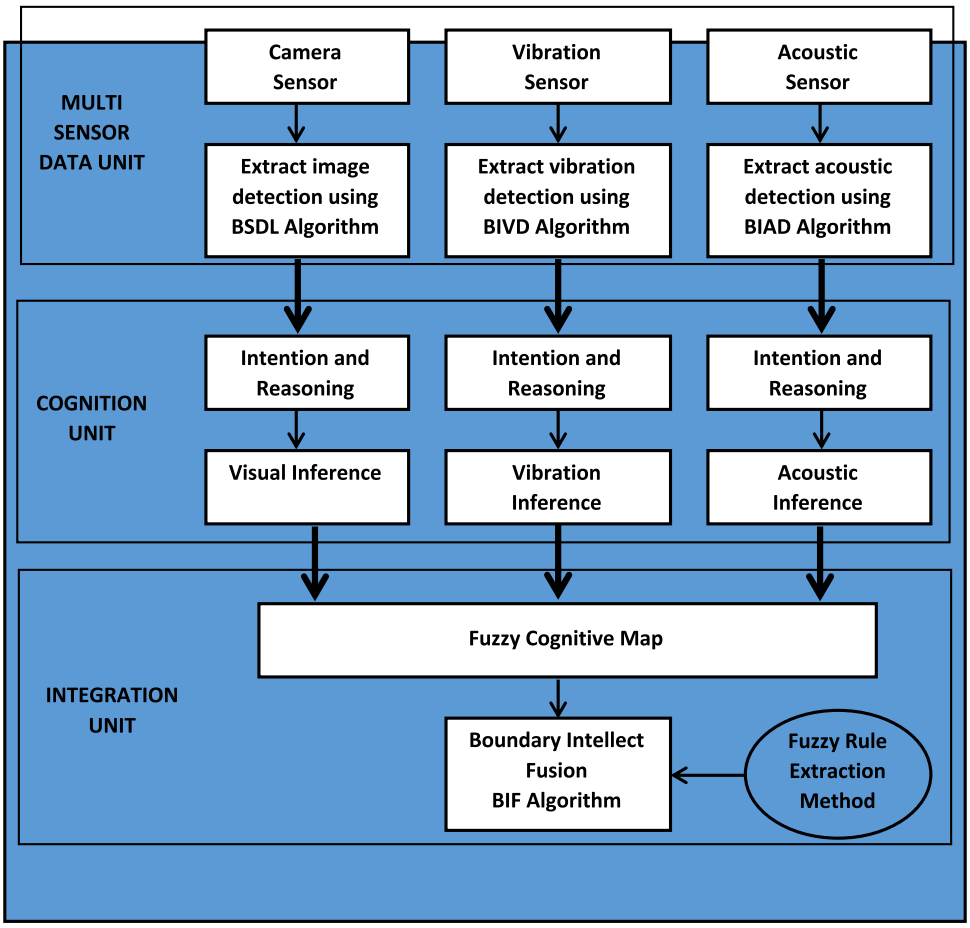


Fig. 1. Taxonomy of boundary intellect system.

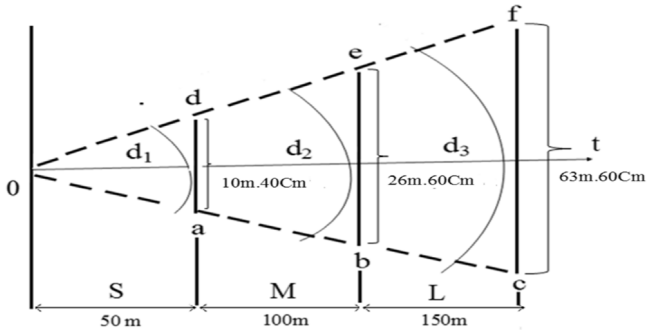


Fig. 2. Distance metric.

for isolating the objects from its background. The idea behind BS is to extract the shape of the entered object in the scene. Third step consists of the elephant image dataset, and includes elephant feature identifying neurons in the multi layered architecture. Classification of the target, training the DCNN network for multi-view elephant dataset consuming gradient descent optimization, feeding in an input image to be predicted and checking the DCNN's prediction for detection rate are the processes involved in this step. The detailed description of this algorithm is discussed in our paper [14].

The Fig. 3 offers the consequence of three diverse ranges. The image is considered as Close-up, when the object comes near to O(n+1). The image is considered as Out-of-focus, when the object goes far from O(n−1).

- Placing the object too close to the camera results in unknown prediction. While testing, only the skin color of elephant is visible in close-up shots.
- The object being too far from the camera, also results in unknown prediction, due to lack of brightness in the camera flash to capture the object (out-of-focus).
- Hence the threshold value is set in three different ranges, namely, short, medium and long to ensure good clarity for the object captured.
  - In the long range, the image appears small, to be considered as O(n−1).
  - In the medium range, the image appears in perfect size, to be considered as O(n).
  - In the short range, the image appears large and is to be considered as O(n+1).

4.1.2. Acoustic sensor

The acoustic sensor used for elephant detection is condenser microphone AKG P120. Boundary Intellect Acoustic Detection (BIAD) steps are proposed in this section for identifying the captured vocalization accurately. The dataset includes 630 recognized vocalizations, in which 125 were not exact, hence 505 vocalizations of elephants were filtered. Four significant steps are proposed. First step includes loading the captured vocalization. The second step includes signal enhancement by removing unwanted disturbances. Feature extraction is performed next, by considering three parameters, fmax, fmin and k. The hearing range is set to maximum (10 000 Hz) and minimum (10 Hz) frequency. K is assumed to be 0.88 for elephants. Greenwood scaled filter is utilized, which consist of 30 bandpass filters. The final step incorporates three

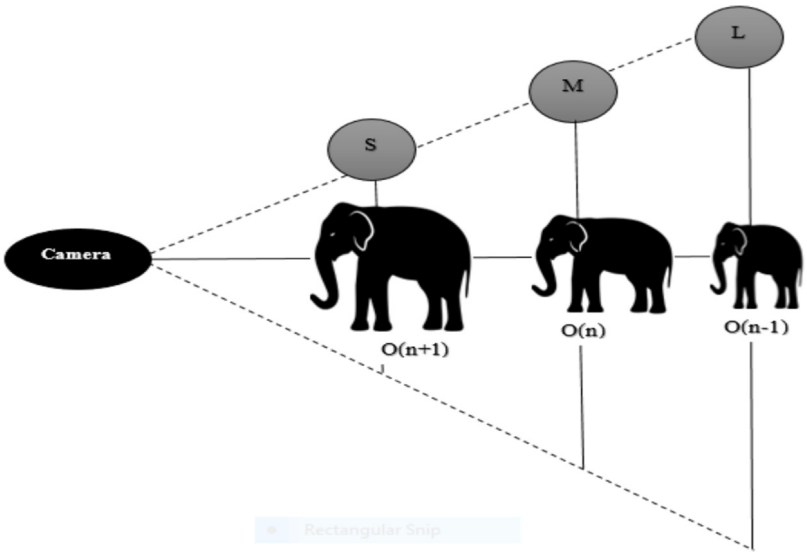


Fig. 3. Distance calculation in camera.

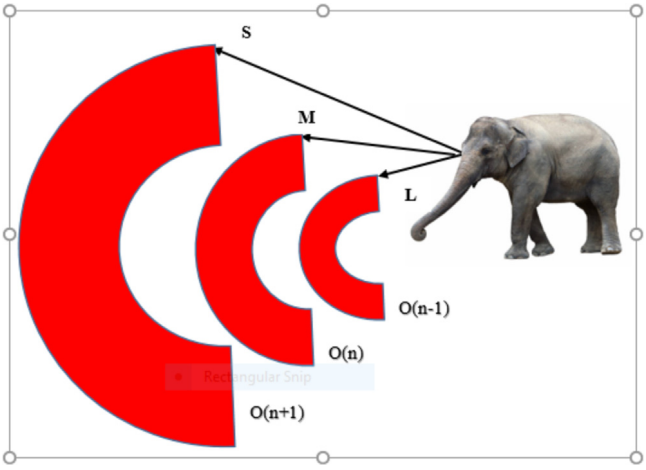


Fig. 4. Distance calculation in microphone.

units, feature aggregation, training and testing. The acoustic detector is trained on the aggregated features and further tested for results [17].

Dissanayake [15] says about the performance variations occurred in the acoustic results due to environmental factors. The major root cause is due to the sound speed dependency on various environmental conditions namely, temperature, water vapor mole fraction, atmospheric pressure and CO_2 content in the air. Out of various factors, temperature play a vital role in sound speed dependency compared to other conditions. Here comes one more significant factor wind, which also modifies the sound propagation. Hence wind and temperature are identified as the major threat for acoustic detection. If these factors are adjusted then, acoustic detection is the appropriate technique for elephant recognition because low frequency calls (rumbles) are the only means used by elephants for long distance communication.

The Fig. 4 offers the consequence of three diverse ranges. Hence the threshold value is set in three different ranges, namely, short, medium and long to ensure good clarity for the vocalization captured.

- Beyond the long range, the vocalization is considered as no target, when the object goes far from O(n−1).
- In the short range, the captured vocalization is considered as Clear, when the object comes near to O(n+1).

4.1.3. Vibration sensor

The vibration sensor used for elephant detection is triaxial MEMS accelerometer. Boundary Intellect Vibration Detection (BIVD) steps are proposed in this section for identifying the captured vibrations accurately. The dataset includes 371 vibrations of elephants collected from different areas. Three significant steps are proposed. First step includes loading the captured vibration and sampling the signal. The second step incorporates fixing of threshold rate. The elephant vibration can be uniquely differentiated from other wild, because of its huge weight. The final step is the determination of biggest and second biggest amplitude, followed by comparison with the fixed threshold rate. If the input vibration exceeds or equal to the threshold, then further detection is performed. Otherwise the input vibration is dropped. According to Geophone distance metric, an array of seismic sensors is placed on the ground following the partitioning of camera. Hence computation followed in previous section of acoustic sensor is followed [21].

4.2. Cognition unit

Elephants have the biggest total mind size of any land creature: Asian elephant −5.5 kg and African savannah elephant — up to 6.5 kg [22]. Bigger minds ought to give more noteworthy data handling power, however the high metabolic expense of cerebrum tissue necessitates that creatures just put resources into substantial minds on the off chance that they have to survive some genuine intellectual difficulties. As a non-expert program, excessively huge to be quite undermined by predators, an elephant's greatest subjective challenge is destined to be social. All types of elephants shape extensive social systems, with progressive, staggered association, suggesting that elephants can manage a level of social intricacy. Adaptable critical thinking would help, and elephants have the vital mental aptitude. It is scarcely astonishing, at that point, that to an ever increasing extent analysts are getting to be keen on the intellectual aptitudes of these perplexing creatures. This cognition unit in Boundary Intellect architecture gives additional information about the target crossing the boundary. The detailed study about elephant cognition is explained in our paper [12]. If no output comes from sensor unit, no information is provided, it means no target found. Based on the sensor output, different cognition feature of elephant can be extracted. For example, Rumble generate from lungs, which ranges from 10 Hz to 500 Hz, when an acoustic sensor results between this range, then the vocalization is categorized as rumble.

4.2.1. Camera

Cognition feature recognition utilizing Elephant images can be categorized as follows, attentive, aggressive, ambivalent, defensive, social integration, mother offspring, sexual, play and death.

- Attentive incorporates listening, looking, sniffing and touching.
- Aggressive incorporates escalation, post conflict, retreat, submissive and threat.
- Ambivalent incorporates snort and croaking.
- Defensive incorporates confront predator, group defense and mobbing.
- Social integration incorporates initiation and leadership, spacial proximity and affiliative.
- Mother offspring incorporates protection and suckling.
- Sexual incorporates attraction, countership, male–male, post copulatory, sexual monitoring and sexual solicitation.
- Play incorporates solicit play, lone and object.
- Death incorporates investigation and intervention.

4.2.2. Vibration

Elephant vibrations can be categories based on their weight, gender and size of the group etc. The experimental setup is organized utilizing an array of sensors for different distance metric, aimed to cover the whole boundary. The vibration dataset includes 371 elephant vibrations. The seismic sensors have 120 m^2 coverage range, when connected as a series.

4.2.3. Acoustic

Identifying Cognition feature using Elephant calls can be categorized based on the origin of noise creation. Laryngeal calls originate from larynx, which incorporates rumble, rev, roar, cry, bark, grunt and husky cry. Trunk calls originate through trunk by a blast of air, which incorporates trumpet, nasal trumpet and snort. Imitated and novel calls incorporates croaking, squelching, chirping, purring and whistling.

4.3. Integration unit

Fuzzy Cognitive Map (FCM) are helpful when the framework is not reasonable to be demonstrated utilizing customary numerical techniques since it is difficult to express the elements or the connection between elements scientifically. FCM is unescapable, this map incorporates each and every components and their connections for creating better and overall complete outcomes, that different methods are not well mapped to the genuine framework. The execution of Boundary Intellect framework incorporates the accompanying strides as appeared in the following Algorithm.

BoundaryIntellect Fusion Algorithm (BIF)

Requirement:

O_{vo} : Ouput Vocalization
O_v : Ouput Vibration
O_i : Ouput Image

Parameters Used:

O	: Output	N	: None
E_e	: Evaluation element	S	: Short
FR	: Fuzzy Rule	M	: Medium
FCM	: Fuzzy Cognitive Map	L	: Long
C1	: None	C2	: Weak
C3	: Moderate	C4	: Strong
C5	: Environment		

Ensure:
Step1
Identify E_e
Load O of all sensors
if (O ≥ Threshold of L)
set O as L
elseif (O ≥ Threshold of M)
set O as M
elseif (O ≥ Threshold of S)
set O as S
else (O = N)
set O as N
endif
Step2
Analyze interrelation of E_e
2a: FR
FR1: if (overall similar O)
then assign same O
endif
FR2: if (dual similar O)
then assign recurrent O
endif
FR3: if (dissimilar O)
then assign C_5
endif
FR4: if (No O)
then assign C_1
endif
2b: FCM
Assign FCM parameters based on three sensor O and FR
C1 = {NNN}
C2 = {LLL, LLM, LLS, LML, MLL, LSL, SLL}
C3 = {LMM, MML, MMM, MMS, MLM, MSM, SMM}
C4 = {SSL, SSM, SSS, LSS, MSS, SMS, SLS}
C5 = {LMS, MLS, LSM, MSL, SML, SML}

Multi sensor data unit inferences three outcomes from the Boundary Intellect framework. In view of the yields of acoustic sensor, camera and vibration sensor basic decision making utilized Fuzzy Cognitive Map (FCM). The Boundary Intellect framework execution is assessed from five perspectives (Strong, Medium, Weak, None and Environment). The Boundary Intellect framework offers three classifications (acoustic sensor, camera and vibration sensor), and every one of these classifications contain their very own components. Rule extraction techniques have been developed. Frequent item set is searched and expert decision is utilized for evaluating concepts. The detailed description of this algorithm is discussed in our paper [12].

5. Data fusion in IoT — boundary intellect detection

This paper implements IoT services to monitor the fixed pseudo boundary in the forest border and provide notification to the forest authority regarding the target entering or crossing the boundary.

As explained in the previous sections the target focused is the elephant, in further the same IoT setup can be implemented anywhere for other wild and human by doing minor alterations in the proposed detection techniques, algorithms and databases. The monitoring pseudo boundary is deployed with multi-sensor setup, here the sensor can be a device designed with sensor (like camera) or individual sensor. The proposed setup utilizes devices with sensors in the multi-sensor unit, namely camera, microphone and geophone. Hence the sensors chosen are camera, acoustic and vibration. The pseudo boundary deployed

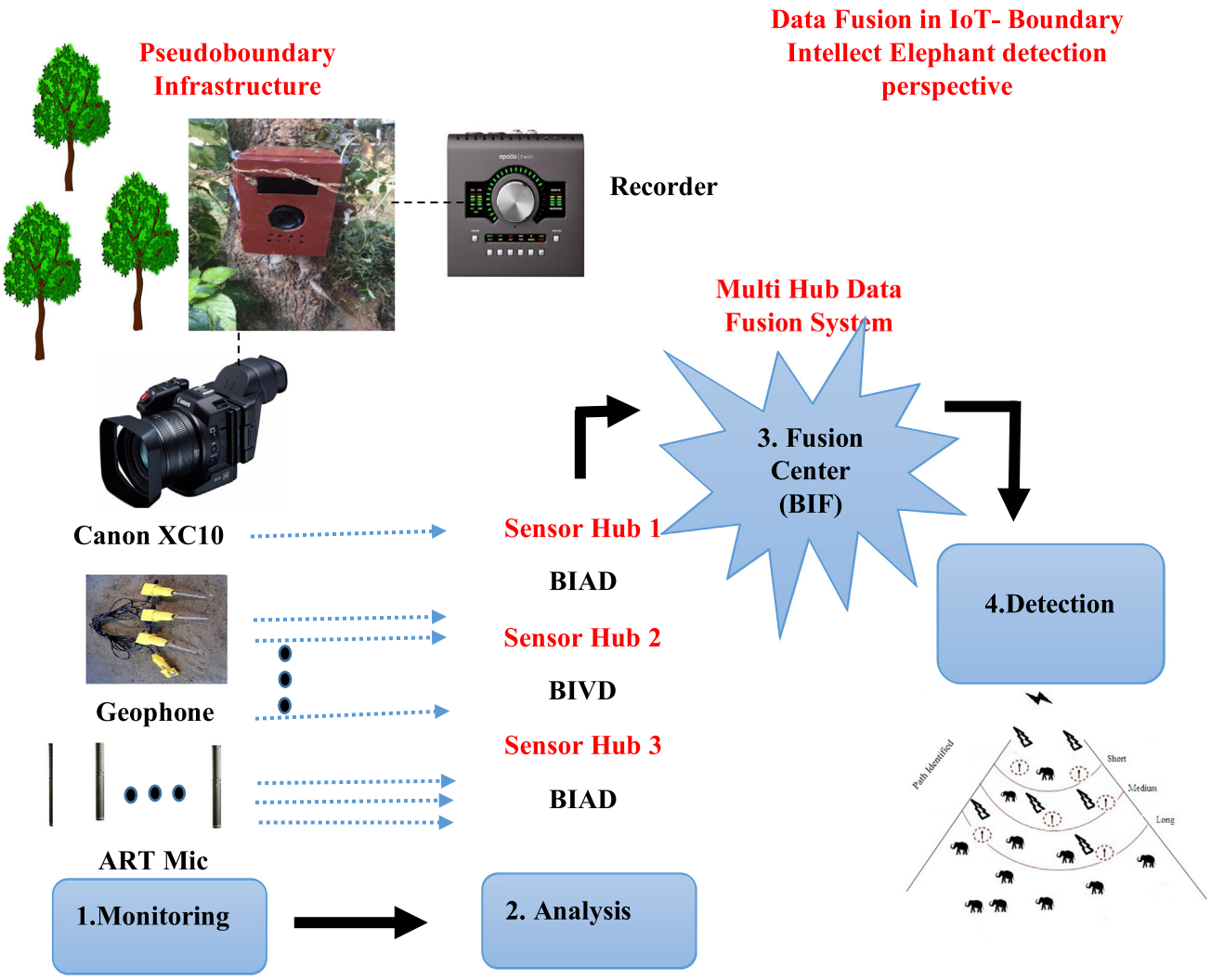


Fig. 5. Architecture diagram of IoT setup.

with WISN setup incorporating with the chosen sensor with devices. The proposed framework incorporates a camera, and an array of microphones and geophones depending on the coverage along with a base station and wildlife observing center. The overall architecture depends on WISNs, which yields effective data gathering and recognition of wild. WISN comprises of many coordinated sensor hubs (mix of image sensor, acoustic sensor and vibration sensor) are conveyed in natural life movement territories or adjacent regions and a sink hub. When a target enters the pseudo boundary, the camera sensor catches picture, vibration signals (Geophone) are recorded in a vibration recorder and the vocalization of the creature recorded through the remote amplifier (Microphone). This three gear data are passed into the sink hub. The three diverse dimensions of qualities are recorded dependent on their separation between IoT unit and the elephant. The separation and the separation count are evaluated using the distance metric and are named as SHORT, MEDIUM and LONG [20]. The architecture shown in Fig. 5 incorporates the overall idea of the previous sections embedded as an IoT application. The novelty of this application is enhanced by integrating the results of different sensor nodes using rule based fuzzy and FCM. Further it takes an attempt to discover cognitive characteristics in elephant. This cognitive feature conveys the reason and intention behind the border crossing target, which is really useful information for the research in this area. Table 1 list the multi sensor and basic components along with its functioning in IoT environment.

Table 1
Components and its functioning.

Components	Function
GSM module	Used to send the message to mobile number stored
Battery	Provides power source to the entire system functioning
Microphone	Used for voice recognition acquired in the field
Geophone	It is used to acquire the vibration for recognition
Wi-Fi	To provide connectivity between devices
Camera	To capture the images of the target object

5.1. System model

5.1.1. Monitoring stage

Multi Sensor Monitoring provides both on-site and out-of-sight detection. Previous sections detailed about the choice of sensors and the steps involved in each computation depending on the nature of sensors. The captured input from different sensors are carried away and processed into their respective hubs.

Three different algorithms were embedded into the microcontroller along with their databases for the comparison process. This monitoring stage is also known as the data collection module, because the sensors collect or extract the features of the target in the sensor unit. Finally, the detection is communicated to the client via the registered mobile number as shown in Fig. 6.

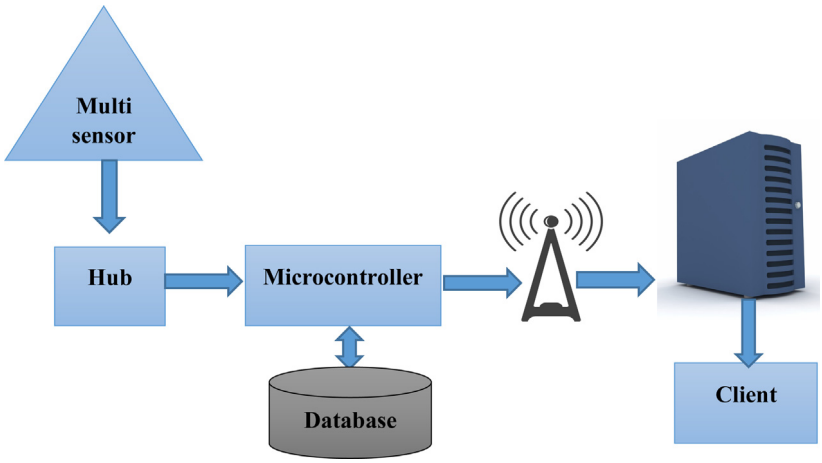


Fig. 6. Monitoring stage.

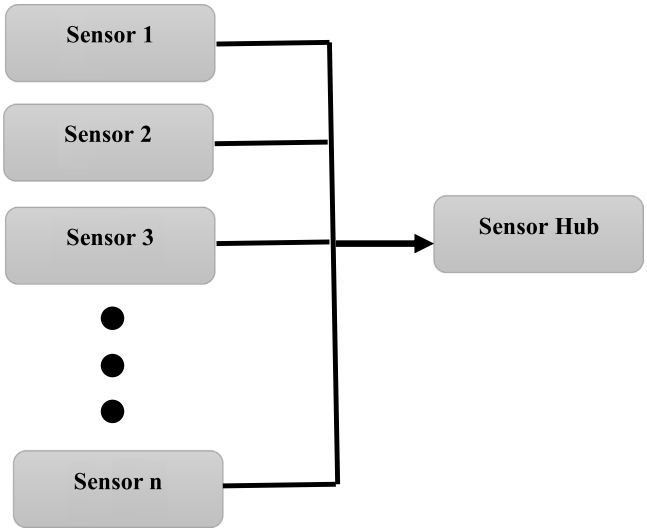


Fig. 7. Analysis stage.

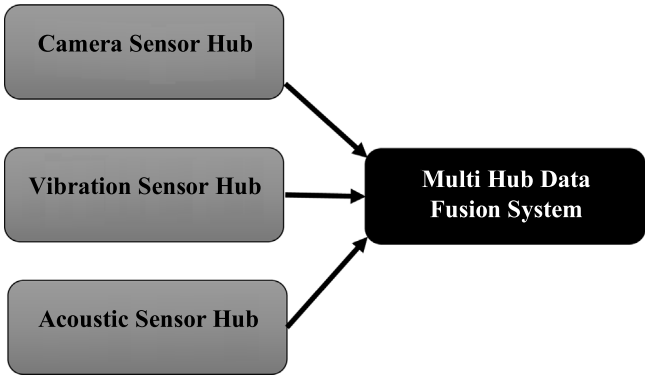


Fig. 8. Fusion center.

5.1.2. Analysis stage

The proposed detection procedures were utilized for analysis of individual sensors. In the IoT environment array of same sensors are also fetched depending on its coverage. Hence multiple captured inputs will be entering the microcontroller.

Though the system has multiple sensors and each sensor can be 1…n, introduction of hub seems to be a good idea for separating different signals depending on sensors nature. The multi sensor unit has three sensor variations; hence the hubs are also classified as three. Each hub is can be holding sensors 1…n of same category as picturized in Fig. 7. After separation of captured signals into the respective hub, respective algorithms were computed and filtrations of unwanted and duplicated inputs were eliminated. The sensors of vibration and acoustic will be in the form of array in the sensor unit.

5.1.3. Fusion center stage

The fusion center is classified into single hub fusion system and multi hub fusion system. Among these multi hub fusion system is highly advantageous and recommended for multi sensor data unit. Here the hubs are classified depending on the nature of sensors. Sensor hub 1 is dedicated for camera, sensor hub 2 for vibration and sensor hub 3 for microphones as shown in Fig. 8. This research is restricted for 3 various detections, hence 3 sensor hubs were initiated. After filtering and evaluating, the final outputs of algorithms will be in the form of distance metric namely, SHORT (S), MEDIUM (M) and LONG (L). Hence multi hub data center stores the output in the form of SSL, SML, LLL etc. Depending on the computational intelligence and the fuzzy rule extraction method, the decision will be made, which is fetched inside the Fusion center with the BIF algorithm, which is also explained in previous section.

5.1.4. Detection stage

The final detection output regarding the target in sensor unit is summarized in Fig. 9. The elephant in the sensor unit is confirmed utilizing the previous steps and the proposed algorithms. Hence this stage after detecting the presence of elephant in sensor unit, informs the client through SMS to the registered mobile number.

5.2. Test bed implementation

Test bed setup (Fig. 10) consists of microcontroller ATmega128, voice recognition sensor, GSM module, microphone, highly equipped camera XC10, H4N handy recorder and Apollo twin duo recorder and battery. Microcontroller incorporates all the above module and gadgets to arrange the general working. Camera XC10 is a high performance camcorder with 4k image quality. Due to its small size and compact nature it is easy to use in this experimental setup. Camera has sensor and image processor which will provide speed for 4k capturing of images with auto focus and image stabilization. Digic DV 5 image processor is present in the capture which enables us to capture the 4k quality images and videos by providing the speed and power. Sensor feature present in the camera will make the camera capture the object

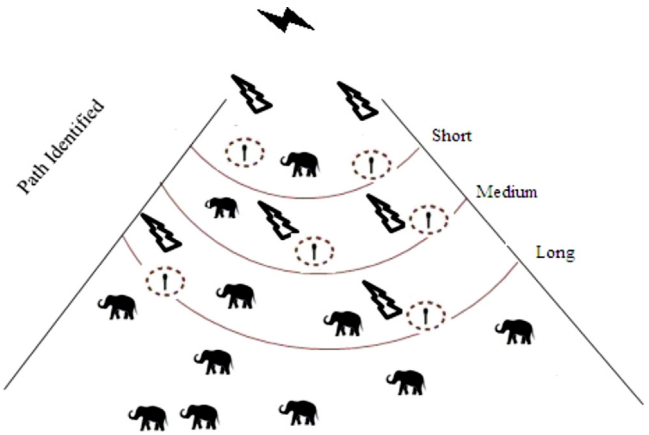


Fig. 9. Detection in sensor unit.

passing through it. Camera is to be placed in metal box fixed in a tree for capturing the images with autofocus mode setting.

Apollo twin recorder is a high class recorder with analog to digital and digital to analog converter in it. The main feature of this device is recording the sound in high quality. It has 2 input channels for connecting microphone. This two input channels can be used for placing the microphones at two different locations connected by the wire. H4N handy recorder is a recorder which offers four tracks of simultaneous recording and an in-built X/Y stereo microphone. The X/Y technique is an excellent way to cover a wide area while still capturing sound sources in the center with clarity and definition which makes it ideal for all types of live stereo recording. Data is received from the multi-sensor unit and further passed to the integration unit. For the entire working of the framework, control is given through the battery backup.

Fig. 11 summarizes the operation of the complete Boundary Intellect System,

- The first module includes vibration sensor, a fixed threshold is compared with the captured amplitudes for Elephant detection.
- The second module consists of acoustic sensor, where the captured vocalization is processed with the mentioned steps for Elephant detection.
- The third module consist of image sensor, which undergoes three important steps namely filtering, cropping and DCNN for predicting the captured image.
- The final module includes the results of all three sensor result and processes them with the framed fuzzy rules and FCM for finalizing the acquired target in the sensor unit is elephant or others.

6. Results

6.1. Multi sensor data unit

The test experiments incorporate 175 elephant images and 25 non-elephant images. BSDL algorithm resulted in 161 positive predictions and 19 negative predictions as shown in Table 2. The experimental setup of camera in the Boundary Intellect sensor unit is shown in Fig. 12(a) along with its coverage. The coverage is divided into three distance metric Short, Medium and Long [12]. The camera angle is well suited for focusing the elephant in medium and long frames. Short results in close-up snaps.

Test experiments were conducted by using Elephant vibration dataset. The experiments include 85 elephant vibrations and 15 other wild and vehicle vibrations. The BIVD algorithm resulted in 78 positive predictions and 10 negative predictions as shown in Table 3. The experimental setup of geophone in the Boundary Intellect sensor unit is shown in Fig. 12(b) along with its coverage. The coverage is divided into three distance metric Short, Medium and Long. The vibration sensor provides 360 degree coverage. Hence an array of geophone is fetched inside the sensor unit.

Test experiments were conducted by using Elephant vocalization dataset. Test experiments include 50 elephant vocalizations and 5 non-elephant vocalizations. BIAD algorithm resulted in 45 positive predictions and 3 negative predictions as shown in Table 4. The experimental setup of microphone in the Boundary Intellect sensor unit is shown in Fig. 12(c) along with its coverage. The coverage is split into three distance metric Short, Medium and Long. Multiple microphones were deployed depending on the coverage range for capturing elephant vocalization inside the sensor unit. The existing approaches were compared using the dataset collected for my research during November 2014 to September 2017, the dataset of elephants utilizing all three sensors were collected on periodic basis from different areas based on their migration nature. Since there is no public benchmark dataset available for evaluation of elephant detector. The comparison tables for computing the detection rate and false alert are as follows.

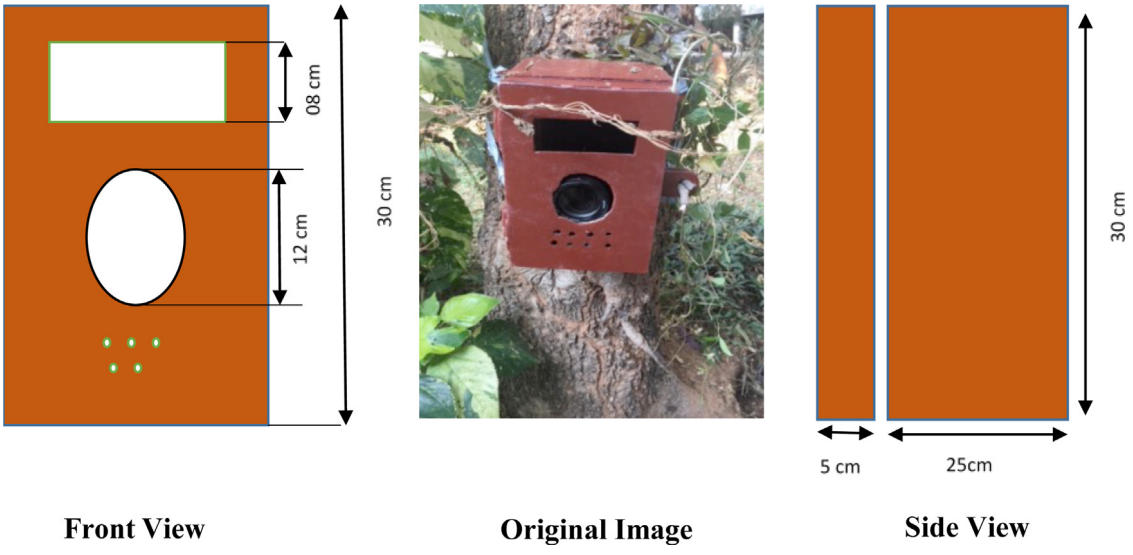


Fig. 10. Specifications of IoT kit.

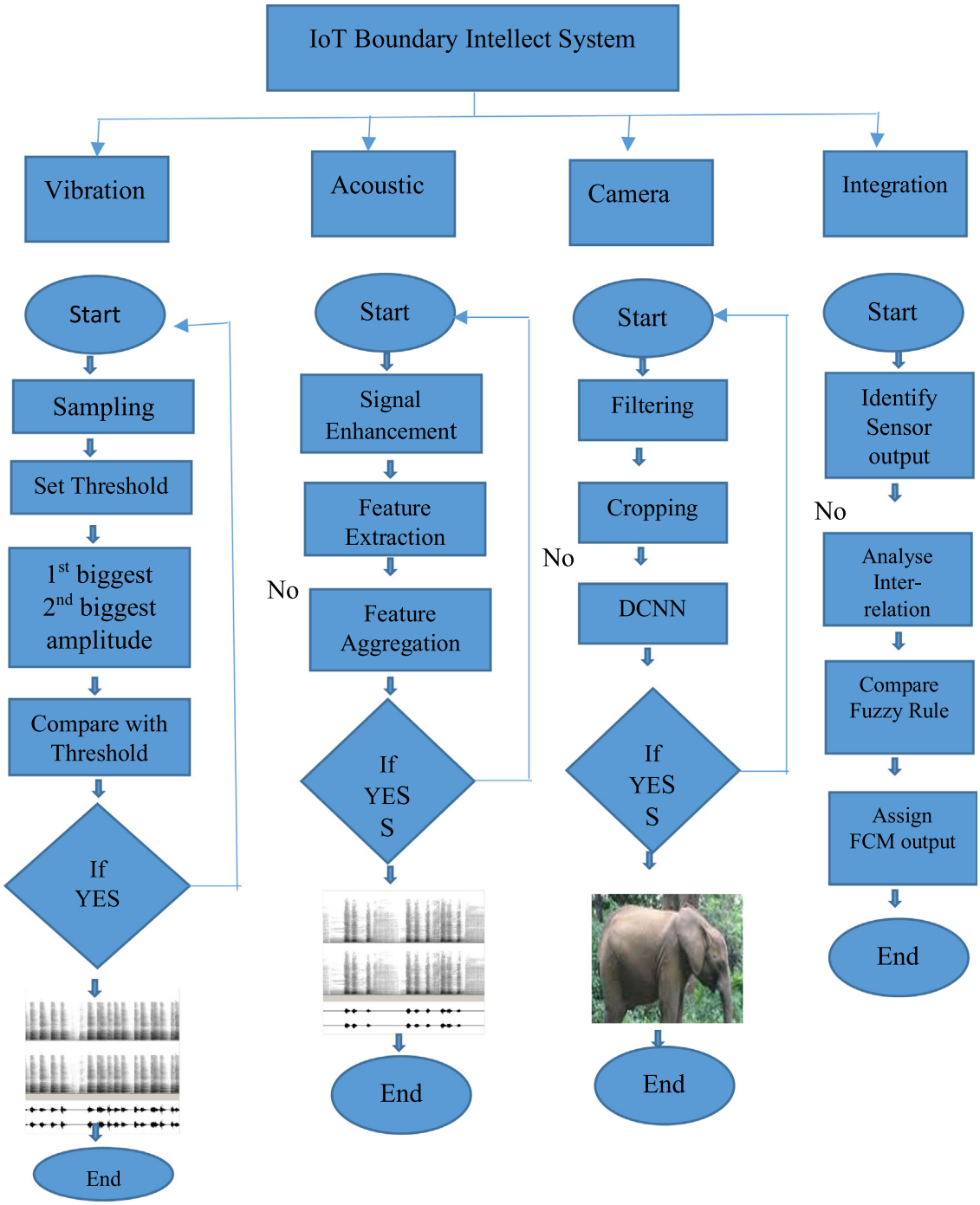


Fig. 11. Complete system operational flowchart.

Table 2
Results of comparison between BSDL and other DL algorithms.

DL algorithms	Accuracy	False positive	False negative
GoogLenet	87.42%	11%	5.5%
GoogLenet + BSDL	89.14%	9.5%	5%
ResNet	89.71%	9%	3%
ResNet + BSDL	92.0%	7%	3%

Table 3
Results of comparison between BIVD and other algorithms.

Algorithms	Positive predictions	Negative prediction	Accuracy	False alert
S.J Sugumar et al. [19]	70	10	82.3%	6.6%
Jason et al. [18]	53	7	62.3%	4.6%
BIVD	78	10	91.7%	6.6%

6.2. BI cognition

The Table 5 summarizes the cognition features identified by our Boundary Intellect (BI) system by utilizing different sensors. Elephant vocalization is classified into various call types and call categories as discussed early. BI acoustic sensor results in the identification of listed cognitive features from the collected dataset. Geophones are capable of collecting vibrations from the sensor unit, and the following

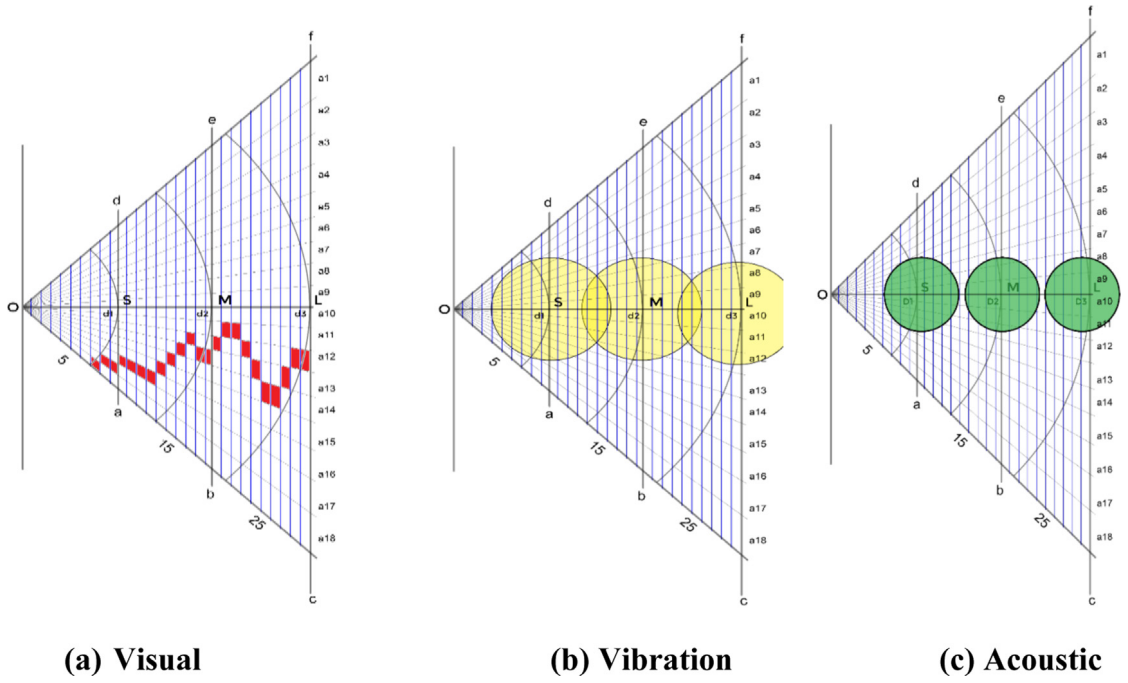


Fig. 12. Coverage of sensors.

Table 4
Results of comparison between BIAD and other algorithms.

Algorithms	Positive predictions	Negative prediction	Accuracy	False alert
Zeppelzauer et al. [17]	45	4	90.0%	1.0%
Hao et al. [16]	39	3	78.0%	3.8%
BIAD	45	3	90.0%	3.8%

cognitive characteristics were identified from the dataset. Cameras play a vital role in identifying cognitive features from elephant images. If no output comes from sensor unit, no information is provided, it means no target found. Based on the sensor output, different cognition feature of elephant can be extracted. For example, Rumble generate from lungs, which ranges from 10 Hz to 500 Hz, when an acoustic sensor results between this range, then the vocalization is categorized as rumble. Utilizing the result of the proposed IoT system (output of individual sensors), Cognitive analysis is made as discussed in Table 5, including cognitive types and categories.

6.3. BIF

In Table 6, performance evaluation is done by utilizing the obtained dataset. Out of this large dataset, only 50 samples are considered for testing. For detailed description and computation refer our paper [12].

Test experiments were conducted by using Elephant dataset. The input supplied during our test experiments for computing various performance metrics are now considered. Test experiments include 50 elephant samples and 10 non-elephant samples. BIF algorithm resulted in 45 positive predictions and 7 negative predictions.

6.4. System analysis

Comparing the performance of VD, AD, ID and AID with BIS, the proposed achieves better results in terms of detection rate and false alert. Both these parameters are controversial, increase in one leads to decrease in another. A successful experimental setup should achieve higher detection rate and lesser false alert. The natural factors around the real time monitoring setup plays a lead role in affecting the performance parameters, because of dynamic environment. Based on these factors, three experimental situations were considered for the

Table 5
Results of cognition unit using multi-sensor output.

S.no	Multi-sensor data	Dataset	Cognition types	Cognition categories
1	Microphone	630 (recognized) 505 (filtered) 125 (not exact)	Laryngeal calls	Rumble Rev Roar
			Trunk calls	Trumpet Nasal trumpet
			Imitated & novel calls	Croaking Squelching Chirping
2	Geophone	371 (recognized)	–	Weight Gender Group size
3	Camera	625(limited for test)	Attentive	Listening Looking Touching
			Aggressive	Escalation Retreat Threat
			Defensive	Confront predator Group defense

Table 6
Results of BIF algorithm.

Sensor unit category	BI-test dataset (test dataset minimized for reducing complexity)	BIF output			
		AS (19)	MS (15)	DS (11)	NT (5)
Visual category Vibration category Acoustic category	50	14.25	7.5	2.75	0

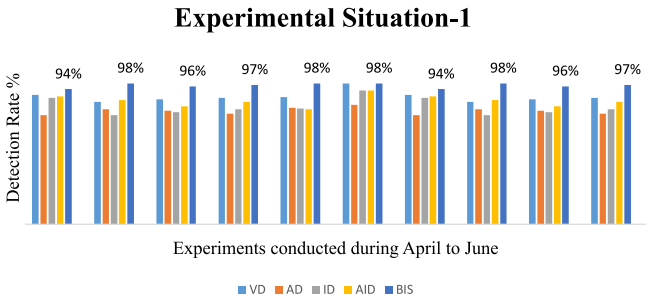


Fig. 13. IoT result-summer.

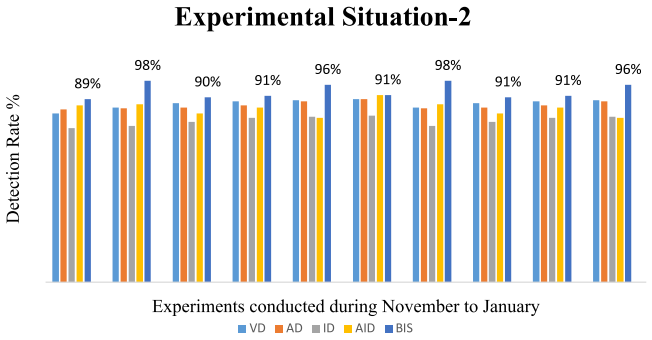


Fig. 14. IoT result-winter.

comparison of the proposed BIS with the existing VD, AD, ID and AID methodologies.

Experimental situation -1: Most of the system works better in sunny environment, until there is no artificial disturbances (vehicle engine sound, interruption of monkeys etc.). The disturbances were man made and rarely animal made. The existing results were better, without considering artificial factors. VD achieves 90% to 98% accuracy with minimal false alert 0% to 1.5%. AD achieves fluctuating results with 76%to 83% accuracy and 8% to 14% false alert. ID accuracy ranges from 88% to 93% with 3% to 5% false alert. AID results were somehow similar to ID, by achieving 89% to 93% with 3% to 4% false alert. With the proposed BIS setup, the results were extremely high in accuracy 94% to 98%, with false alert equals to 0% to 1.5%. To the evaluation VD achieves better results when compared to other existing methodologies in summer. The proposed setup reaches remarkable accuracy and false alert, hence suitable for sunny environment as shown in Fig. 13.

Experimental situation — 2: During winter, mid-morning results were clear, when compared to early morning, because of fog. The result of VD and ID are dropped, when compared to AD, AID and BIS. VD obtained result ranging from 82% to 89% with false alert 3% to 5%. ID pictures where not clear, achieving accuracy 75% to 81% and false alert 6% to 8%. AD results where improved when compared to above, acquiring detection rate 84% to 89% with 3% to 6% false prediction. AID reaches better results similar to AD ranging 86% to 91% with 3% to 5% false prediction. The proposed setup acquires good results with detection rate 89% to 91% with 2% to 5% false prediction. To the evaluation AD and AID achieves better in winter, when compared to others. The proposed BIS achieves comparatively improved results, hence works for winter climate too as shown in Fig. 14.

Experimental situation — 3: The results were poor during rainy season, because of wind and rain, when compared to all other climates. VD obtained 60% to 64% accuracy and false alert 6% to 12%. AD accuracy ranges from 62% to 70% and false prediction 5% to 8%. ID acquired 75% to 82% detection rate and false alert 3% to 6%. AID produces 77% to 85% accuracy, with 2% to 5% false alert. The

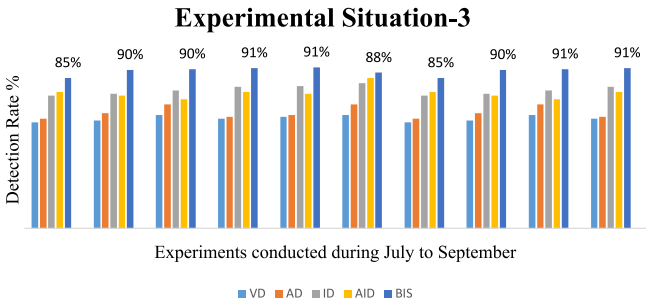


Fig. 15. IoT result-rainy.

Table 7
Results of IoT.

Algorithms	Accuracy	False alert
BSDL	87.34	4.5
BIVD	88.65	2.9
BIAD	70.4	14.0
BIF	91.1	2.2

proposed setup is efficient enough when compared to the existing, while achieving 85% to 88% accuracy and false alert ranging from 2% to 5%. Hence suitable even in rainy season as shown in Fig. 15.

Table 7 summarizes the overall result considering all climatic conditions from IoT kit fetched in real forest border in Hosur, Tamilnadu. BIDL and BIVD algorithms achieves remarkable performance metrics when fetched in real time environment. The major challenge lies with BIAD algorithm, where the input for acoustic setup became rare, if obtained the noise and other disturbances played a vital role. BIF algorithm follows the fuzzy rule and produced the integrated result with minimized false alert. Overall the real time monitoring IoT system produce successful output achieving significant performance metric in all climatic conditions.

The following is the computation followed for evaluation of accuracy and false alert,

- Accuracy can be computed as follows,

$$\text{Accuracy} = \frac{\#\text{ of obtained positive predictions}}{\#\text{ of true positive}}$$

- False positive can be computed as,

$$\text{Fpos} = \frac{\text{variation between obtained positive predictions and true positive}}{\text{total number}}$$

- False negative can be computed as,

$$\text{Fneg} = \frac{\text{variation between obtained negative predictions and true negative}}{\text{total number}}$$

7. Performance evaluation

7.1. Real time setup

The real time monitoring elephant detector is implemented for the proposed Boundary Intellect framework and the same is compared with the recent works. The articles taken for comparison includes all categories of detection. In single level detection two categories are considered, detection with geophone by S.J. Sugumar [19] and detection with microphone by C.M. Dissanayake [15,23]. In two level detection the only trial made is with the combination of above ground sensors, detection with microphone and camera by Zeppelzauer [17]. There is currently no public benchmark dataset for the evaluation of elephant detectors. However, we can compare the results of the proposed approach with that of related approaches [15,17,23] and [19]. An objective comparison with the results of [19] is not appropriate, since

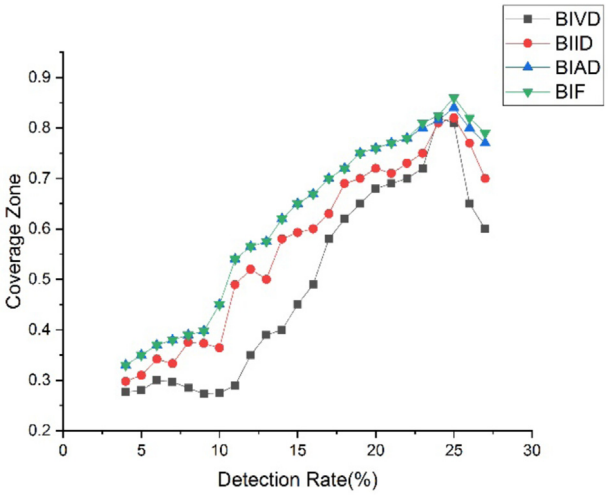


Fig. 16. Detection rate.

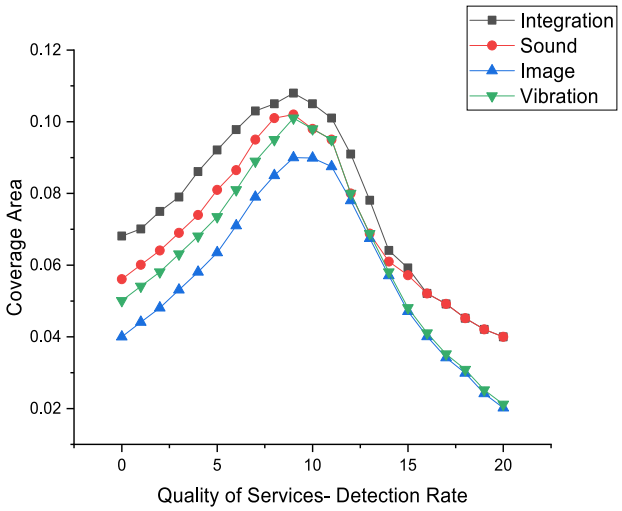


Fig. 17. QoS.

the author uses a dataset for few minutes of evaluation. [15,17,23] comprises few hour recording, the vocalization and images obtained is less. Our recording continuously last for 55 h, hence resulted in huge dataset.

The following are the real-time setup parameters considered for performance evaluation.

1. **Number of positive prediction:**
   The positive prediction from the sensor unit varies depending on the sensor/ the complete device deployed. Considering the comparative study of the above articles positive prediction is not consistent, it varies depending on the nature of climatic conditions for above ground sensor and soil humidity for the underground sensor. The same occurs in the proposed setup, but the incorporation of both the combination of sensors mutually compensate their weakness. Hence the number of positive prediction increases in the proposed setup as shown in Fig. 16.
2. **Number of false alert:**
   False prediction is the major reason behind the unpopularity of technical setup in elephant detection in wild environment. The number is massively limited in the proposed setup when compared to others. Because of not considering the single sensor failure, by following the fuzzy rule.
3. **Efficacy of integration and cognition:**
   The core idea of integration and cognition increases the novelty of the proposed setup, which cannot be compared with any. The success percentage lies behind the performance evaluation is based on integration. Cognition predicts the reason and intention behind the boundary crossing of the target, along with the additional information of the creature.
4. **Anonymity threshold value:**
   Additionally, the proposed framework performance is analyzed by means of anonymity threshold value. SHORT, MEDIUM, HIGH are the parameters supplied for positive prediction and ENVIRONMENT, NO TARGET are the parameters utilized for negative prediction. The threshold value set for these parameters varies depending on the area size and the group size of the target entering the pseudo boundary.

7.2. Performance metrics

In addition to the performance bounds mentioned in the previous section, three performance metrics are defined to measure the effectiveness of the proposed framework. The performance metrics listed as follows,

1. Quality of Service (QoS)
2. Performance (P)
3. Detection rate guarantee (DRG)

Fig. 17 shows the test results of QoS. The graph represents the variation happening in the predictions of individual sensors. The increase in the positive predictions and decrease in the false prediction, improves the QoS. This tendency of improving the QoS continued until it becomes almost consistent. The performance of the proposed setup merely lies on the consistent result of the QoS. Hence these metrics P and QoS are directly depending on each other. To avoid the conquest in the prediction DRG is presented. This metric is the ratio of the number of true prediction against false alert. The percentage of final integrated algorithm output is consistently increased, when compared to individual sensor results. This increases the DRG of the proposed framework compared to the listed articles.

8. Discussion

The simulated detection results in 97.14% detection rate and 2.5% false alert and the real time test bed result in 91.1% detection rate and 2.2% false alert. A test bed result come across various disturbances and rarely gives acoustic output. Most of the time it produces two active sensor outputs namely vibration and camera. Figs. 18 and 19 presents the positive prediction of two sensors and three sensor of the proposed framework. Each sensor has its own merits and demerits. Acoustic detection enables out of sight target recognition. But highly challenging because vocalizations are of low frequency and captured with multiple noise sources. Vibration recognition is best suitable for species classification. But rarely fails because of an effect of climate conditions and soil humidity. Image detection acts as the visual proof for elephant detection, but becomes challenging because of reduced lightings and backgrounds. An integrated detection is truly promising because each technique mutually compensates their weaknesses, leading to a more reliable detection [12]. The benchmark elephant detection articles taken for comparison are described in previous section. The performance evaluation process discusses the reason behind the increase of performance metric namely QoS, P and DRG. The novelty of the proposed framework is highly enhanced based on integration and cognition. Table 8 list the performance metric according to the distance metric in sensor unit.

Climatic conditions play a vital role in achieving accuracy and false alert. Comparison with the existing study proved that the proposed system is well suited for all climates. Experimental results acquired

Table 8
Results of distance metric in algorithms.

Multi sensor	Parameters	Distance metric		
		S	M	L
Camera BSDL	Clarity	Closeup	Accurate	Focusable
	Accuracy	0.5%	97.4%	94.6%
	False Rate	10.3%	0%	2.9%
	Cognition	✘	✔	✔
Vibration BIVD	Clarity	Good	Good	Good
	Accuracy	82.4%	96.5%	98.4%
	False rate	0%	0%	0%
	Cognition	✔	✔	✔
Acoustic BIAD	Clarity	Good	Unpredictable	Unpredictable
	Accuracy	86.8%	60.5%	42.9%
	False rate	12.5%	38.7%	53.1%
	Cognition	✔	✘	✘
Integration BIF	Clarity	Good	Accurate	Good
	Accuracy	93.2%	100%	97.8%
	False rate	2.5%	0%	1.2%
	Cognition	✔	✔	✔

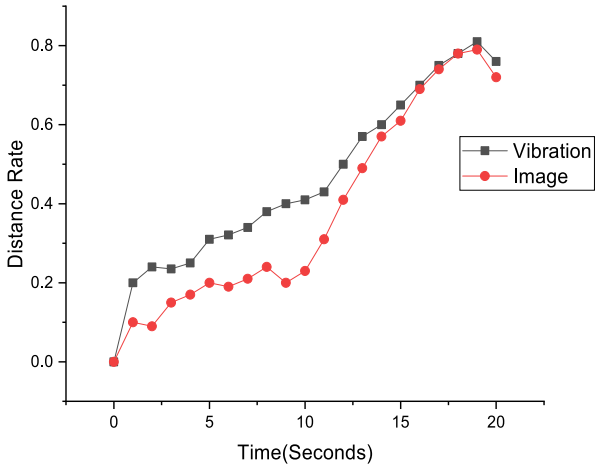


Fig. 18. Prediction from dual sensor (DS).

finest results for the proposed BIS. Single level detection achieves better results based on particular climate for example, VD provides results in summer, but not in rainy. Two level detection in the combination of both aboveground sensors provides similar results as discussed in result section. Hence does not make much difference in achieving detection rate and false prediction.

This research is the attempt considering all factors, is the possible combination of sensors with underground and aboveground with integration of fuzzy rule extraction and FCM. The failure of some sensors is not affecting the output, because of integration. Hence achieving better output in all climatic conditions, along with cognitive features. Cognition helps in understanding the completely about the creature crossing the boundary.

Table 9 compares recent existing works with the proposed prototype. The novelty of our proposed work is enhanced by three level detection along with technical advancements. Recent works were taken as benchmark for validating the proposed prototype. Communication range is enhanced based on the integration of sensors. Vibration detection (VD) [19] does not have more works; the major limitation is because of the soil humidity happening underground. Acoustic detection (AD) [15] is the recent work done for acoustic detection of elephant, which is the very rare phenomenon happened in our test bed and its challenging too. Image detection (ID) [9] is highly popular, which gives the visual proof for the automated detector systems. Two level detection [17] combines both acoustic and image for elephant detection (AID), which incorporates only above ground sensors. The

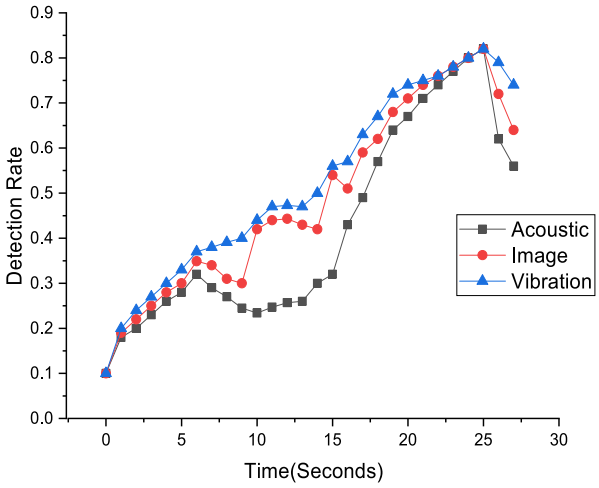


Fig. 19. Prediction from all sensor (AS).

proposed Boundary Intellect System (BIS) novelty is highly enhanced based on the combination of underground and above ground sensors, which resulted in three level detection. The integration part is too complicated and improves the detection rate of real time monitoring system.

9. Conclusion

In this paper, Boundary Intellect is proposed and implemented through Wireless Integrated Sensor Network architecture for elephant detection by improving the detection rate and achieving minimized false alerts. A hardware platform of IoT design is implemented for real time monitoring of elephant detection in forest boundary. BI integrates three level of detection along with cognition and computational intelligence techniques. Finally, the system is deployed in Hosur forest region, Tamilnadu, India. Further the real time implementation successfully identifies target entering or crossing boundary within 50 m accuracy in the pseudo boundary setup in the real forest with remarkable accuracy. Cognition features identified in the real time test bed resulted in migration due to climatic changes and always in group. Acoustic cognitive feature is very challenging to recognize. The main advantage of this approach is its simple structure and flexibility, representing knowledge visually and more descriptively. Hence, it might be a convenient tool in predicting elephant intrusion in forest borders and minimizing Human–Elephant Conflict.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Arun Kumar Sangaiah: Conceptualization, Methodology, Validation, Investigation, Data curation, Visualization, Writing - original draft, Writing - review & editing, Supervision, Project administration, Formal analysis. **Jerline Sheebha Anni Dhanaraj:** Software, Validation, Investigation, Resources, Data curation, Visualization, Writing - review & editing. **Prabu Mohandas:** Methodology, Software, Resources, Validation, Writing - review & editing, Data curation. **Aniello Castiglione:** Conceptualization, Methodology, Validation, Writing - review & editing, Investigation, Supervision.

Table 9
Comparison of our proposed prototype with existing devices.

	VD	AD	ID	AID	BIS (Proposed)
Parameters	1. Vibration 2. Remaining capacity 3. Full charge capacity 4. Voltage	1. Acoustic 2. Remaining capacity 3. Full charge capacity 4. Voltage	1. Camera 2. Remaining capacity 3. Full charge capacity 4. Voltage	1. Acoustic 2. Camera 3. Remaining capacity 4. Full charge capacity 5. Voltage	1. Battery ID 2. Acoustic 3. Vibration 4. Camera 5. Remaining capacity 6. Full charge capacity 7. Voltage 8. State of charge 9. Average current
Level sensing	1-level detection	1-level detection	1-level detection	2-level detection	3-level detection
Communication protocol	Bluetooth	Wi-Fi	Wired	Wired	Wi-Fi
Communication range	5 m (outdoor)	30 m	Based on the length of the wire	Based on the length of the wire	50 m (outdoor)
Battery types	12 V	12–48 V	12–48 V	12–48 V	12–48 V

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